

3.3.7 Wheels

Wheels when attached to the axle of motor provide locomotion. There are certain aspects of wheel you should consider before buying them:

Wheel texture

The texture of the wheel decides how much grip your wheel would have on the surface it moves upon. If the surface is very smooth, then you need to have wheels which have rubber coating on them, so that they can get good grip. If the surface itself is too rough, you can afford to use plastic wheels. However, it's always better to use rubber coated wheels as then you don't need to worry much about the surface.

Wheel diameter

Wheels with large diameter give you lower torque but higher velocity. If you have a motor with high torque, you can make use of high diameter wheels. Also, high diameter wheels are required when you need large ground clearance, which is defined as the height of the base of your robot from the ground. So if you plan to make an 'All Terrain Type' of robot, use rubber coated wheels with big wheels. However, if you don't need too much ground clearance you can use small wheels in your robot. A point to keep in mind is that the diameter of the wheel must always be bigger (by at least 3-4 cm) than the diameter of the motor, and if they are equal or nearly equal, the ground clearance would be zero and the base of the robot would touch the ground.

Wheel center hole diameter

The center hole is the slot where the shaft of the motor would be inserted. So, the diameter of this slot in your wheel must be greater than the diameter of the motor shaft. You can still manage if the wheel center hole is too big for a shaft, as then you can put in some paper or glue to jam them together, but if the slot in the wheel is smaller than the motor shaft, the shaft would simply not enter it. 